

RAG Evaluation: Metrics, Testing & Excellence

A practical guide to measuring, testing, and continuously improving Retrieval-Augmented Generation systems — from retrieval quality to end-to-end pipeline performance.

ML ENGINEERING

RAG SYSTEMS

EVALUATION

Retrieval Metrics: Measuring What Gets Found

Before the LLM generates anything, the retriever must surface the right context. These four metrics quantify retrieval quality at different levels of granularity.

Precision

Of retrieved documents, what fraction are relevant?
Penalizes noisy retrievals.

$$\text{Precision} = \frac{\text{Relevant Documents Retrieved}}{\text{Total Documents Retrieved}}$$

Recall

Of all relevant documents, what fraction were retrieved?
Measures coverage.

$$\text{Recall} = \frac{\text{Relevant Documents Retrieved}}{\text{Total Relevant Documents in Database}}$$

MRR

Mean Reciprocal Rank — rewards systems that rank the first relevant result higher.

- **What it measures:** How early in the ranked list the *first* relevant document appears.

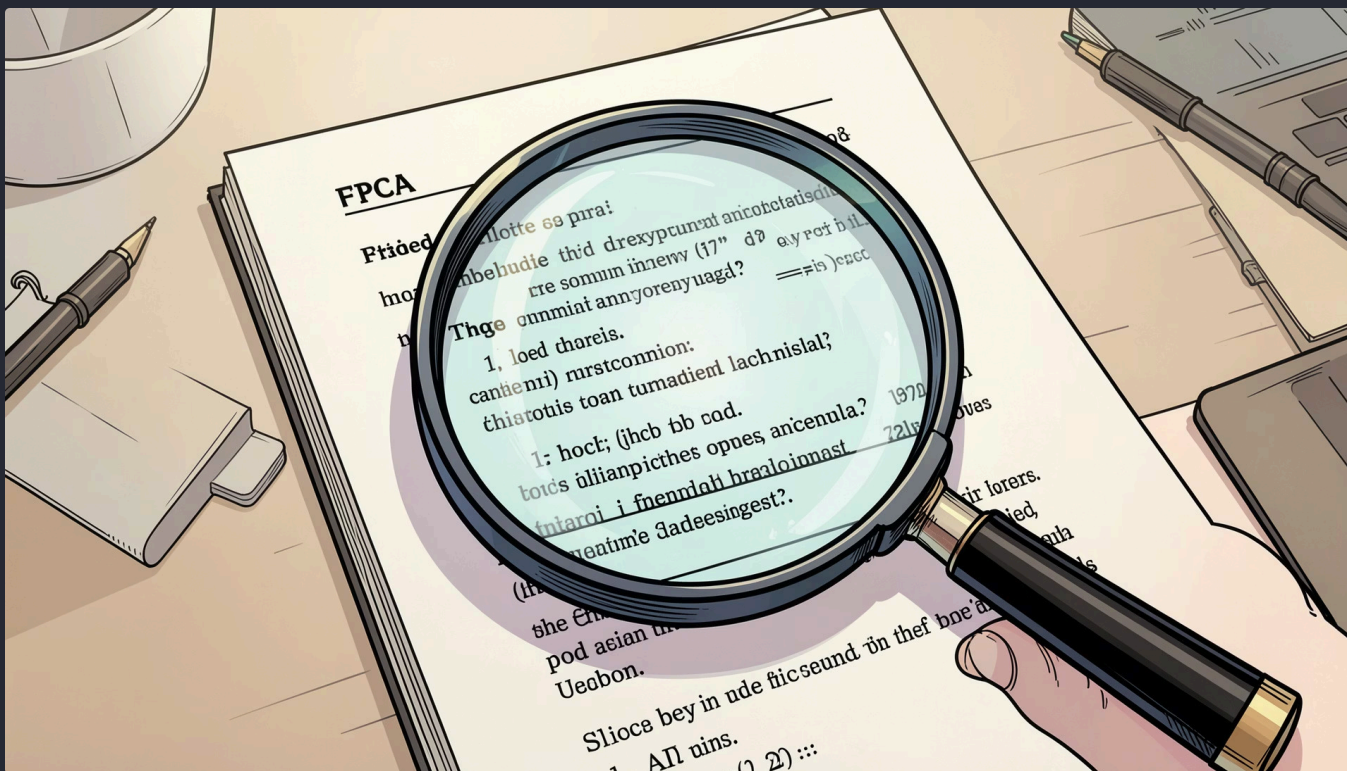
$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$

NDCG

Normalized Discounted Cumulative Gain — accounts for graded relevance across ranked positions.

What it measures: The overall ranking quality. Unlike other metrics, it supports **graded relevance** (e.g., scoring documents as 1, 2, or 3 based on how *good* they are) and penalizes relevant documents that appear too far down the list

Generation Metrics: Faithfulness & Relevance



Faithfulness

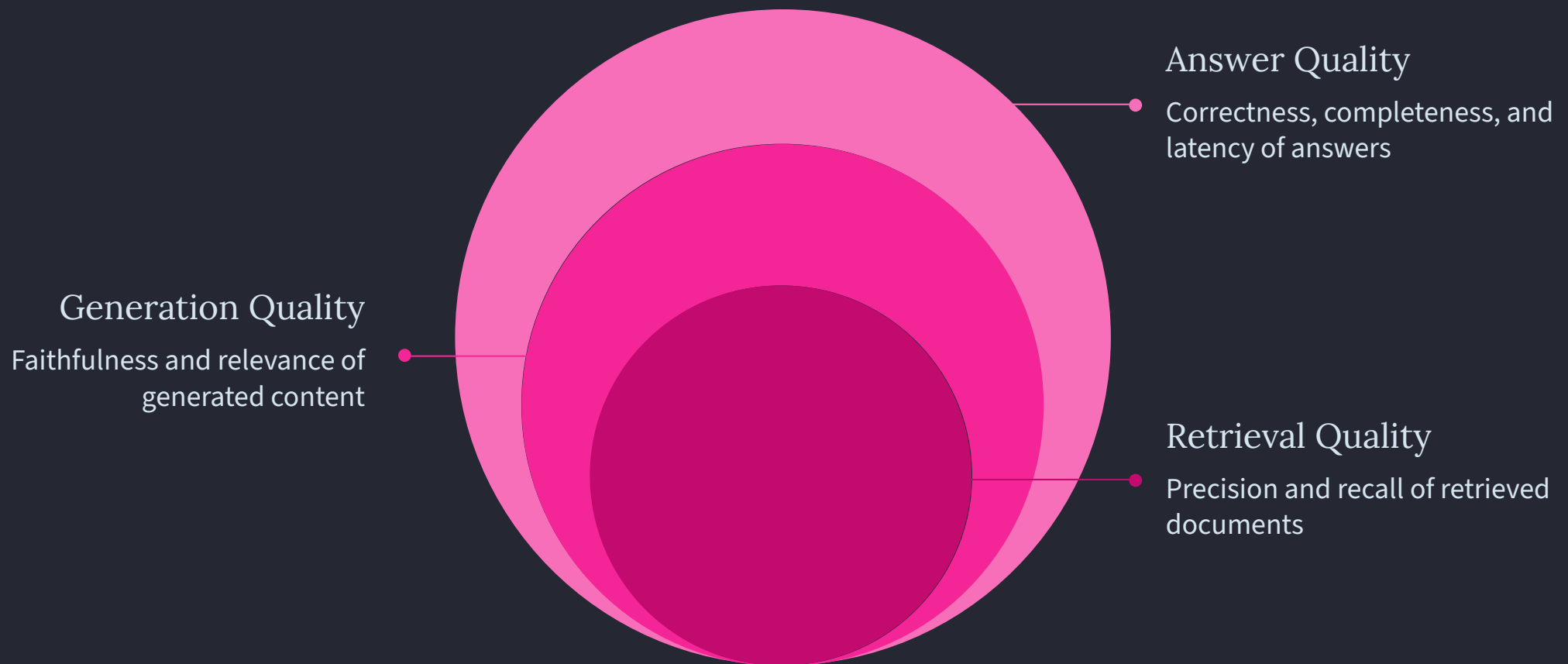
Does the generated answer stay grounded in the retrieved context? Faithfulness detects hallucinations — claims not supported by source documents. Measured via NLI models or LLM-as-a-judge.

Relevance

Does the answer actually address the user's query? Relevance is orthogonal to faithfulness — an answer can be faithful yet irrelevant. Typically scored 0–1 by an LLM evaluator.

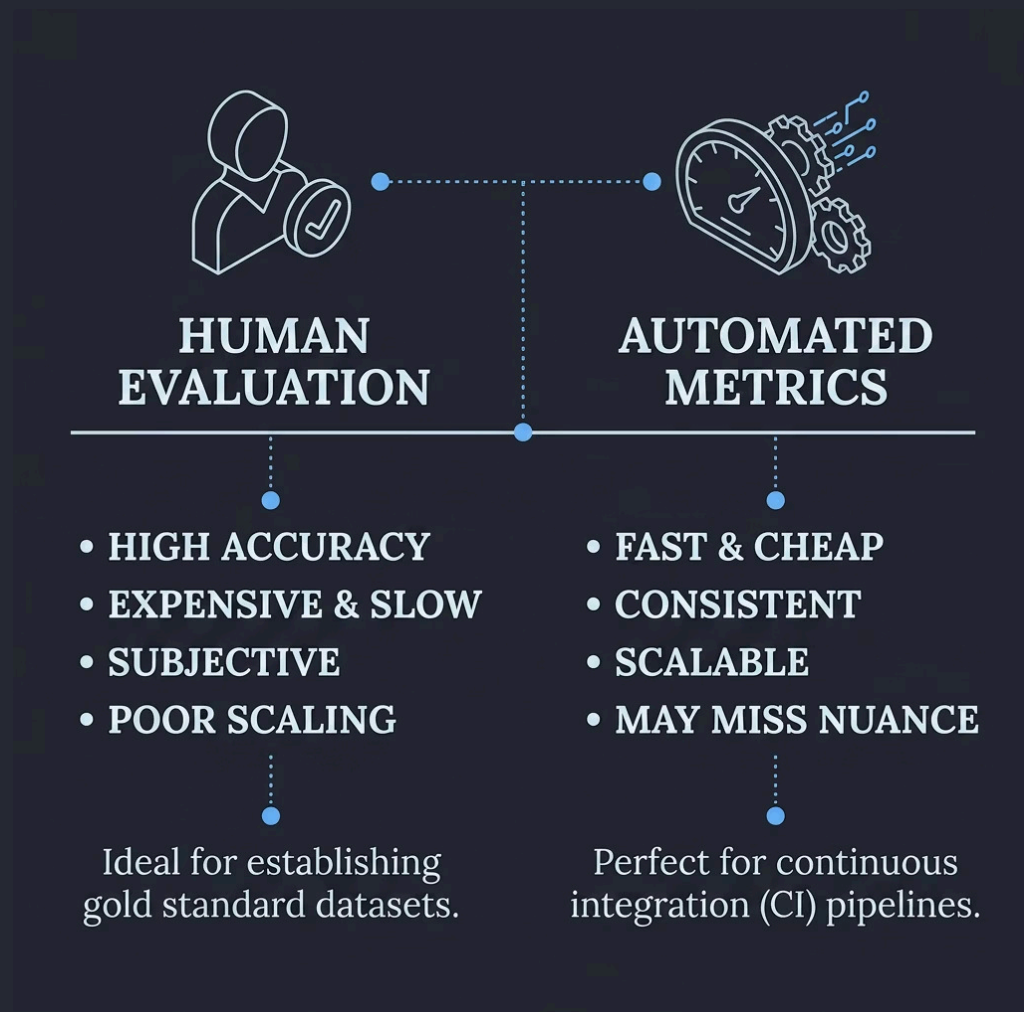
End-to-End RAG Metrics

Individual component metrics are necessary but insufficient. End-to-end evaluation captures the full pipeline's real-world effectiveness.



The best-performing retriever does not always yield the best final answer. Track **Answer Correctness** (semantic similarity to ground truth) and **Context Precision** (relevant chunks ranked above irrelevant ones) as your primary E2E signals.

Human Evaluation vs Automated Metrics



When to Use Which

- **Human eval** for gold test sets, regulatory review, and edge-case validation
- **Automated metrics** for regression testing, A/B experiments, and nightly pipelines
- Hybrid approach: use LLM-as-a-judge calibrated against human labels

i Calibrate your LLM judge monthly against fresh human labels to prevent evaluator drift.

What is Evaluator Drift?

Over time, the behavior of an LLM judge may become misaligned with human expectations.

Reasons include:

- Model updates
- New domains and data
- Changes in user behavior
- Prompt modifications
- Distribution shifts in queries

For example:

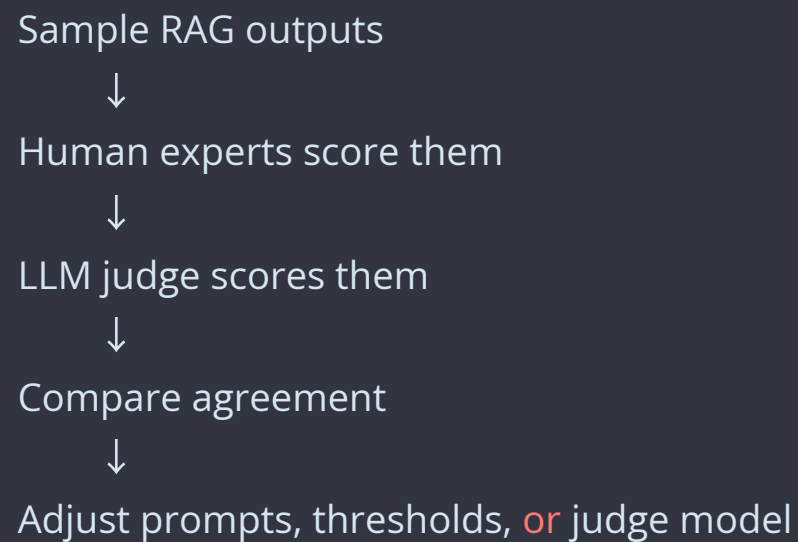
Time	Human Score	LLM Judge Score
January	9	9
June	9	6
September	9	5

The judge is "drifting" away from human judgment.

What does calibration mean?

Calibration means comparing the LLM judge's ratings against **fresh human-labeled examples**.

Process:



You might compute:

- Correlation with human ratings
- Agreement percentage
- Cohen's kappa
- Precision/recall on pass/fail decisions

Why monthly?

A monthly cadence is common because:

- Production data changes continuously.
- New document collections are added.
- User queries evolve.
- LLM providers may update models.

LangSmith: Building Comprehensive Test Datasets

1

Seed with Real Queries

Start from production logs — real user queries capture distributional complexity synthetic data misses.

2

Augment with Variants

Use paraphrasing, typos, multilingual variants, and adversarial rewrites to expand coverage.

3

Annotate Richly

Tag each example with expected answer, relevant chunk IDs, difficulty level, and domain category.

4

Version Your Dataset

LangSmith datasets are versioned — track changes alongside your RAG pipeline versions for reproducibility.



LangSmith Evaluators: Configuration & Customisation

Built-in Evaluators

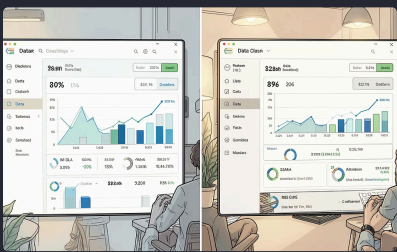
- **LLM-as-a-judge** — configurable prompts for faithfulness, relevance, correctness
- **String match** — exact, regex, or semantic similarity thresholds
- **Critique evaluators** — structured rubric-based scoring

Custom Banking Domain Evaluators

- Compliance check: flag responses missing required disclosures
- Product accuracy: validate interest rates, fees against source docs
- Tone guard: ensure regulatory-appropriate language

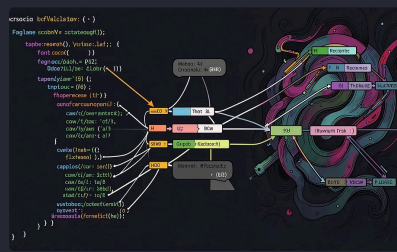


A/B Testing & Regression for RAG



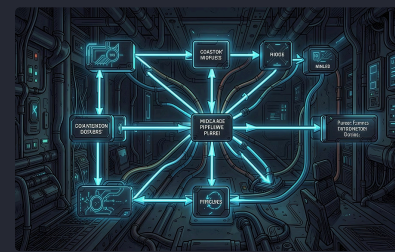
A/B Testing Configurations

Compare chunk sizes, embedding models, or LLM temperatures in LangSmith. Route a percentage of eval queries to each variant and measure statistical significance across metrics.



Regression Testing

Run your full eval dataset against every pipeline change. Flag regressions where faithfulness or recall drops beyond a defined threshold. Block deployments on critical metric degradation.



Continuous Evaluation Pipelines

Trigger LangSmith eval runs on every merge. Integrate with CI/CD to gate releases. Schedule nightly full-suite runs to catch drift from index or model updates.

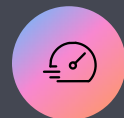


Advanced RAG Testing: Adversarial & Stress



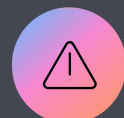
Adversarial Testing

Inject prompt injections, conflicting documents, and ambiguous queries to probe robustness boundaries.



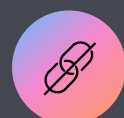
Stress Testing

Simulate high query volume to measure latency degradation, rate-limit behaviour, and vector DB throughput.



Edge Cases

Test empty retrieval, multi-intent queries, out-of-domain questions, and extremely long context windows.



Failure Mode Analysis

Classify failures by root cause — retrieval miss, chunk truncation, LLM misinterpretation — and track frequency over time.

Cost-Performance Tradeoffs & Key Takeaways



What the Data Shows

Diminishing returns set in quickly. A strong retriever with a mid-tier LLM often outperforms a weak retriever paired with the best model — at a fraction of the cost.

Your Evaluation Checklist

- Track retrieval + generation + E2E metrics separately
- Use human labels to calibrate LLM judges
- Run regression tests on every pipeline change
- Stress-test before production deployment
- Optimise for cost-quality frontier, not peak quality alone